Logistic Regression

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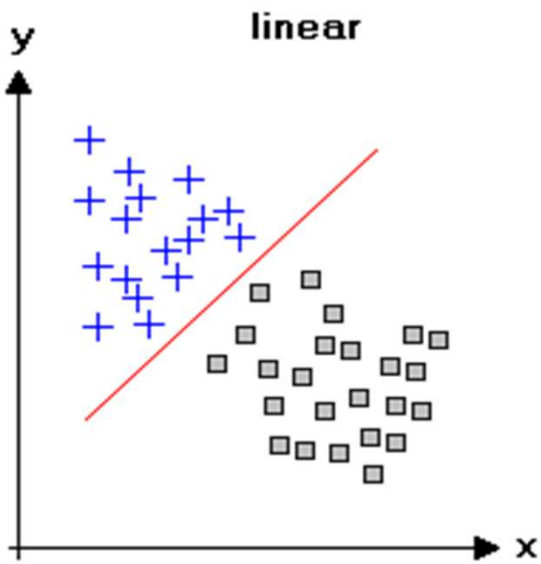
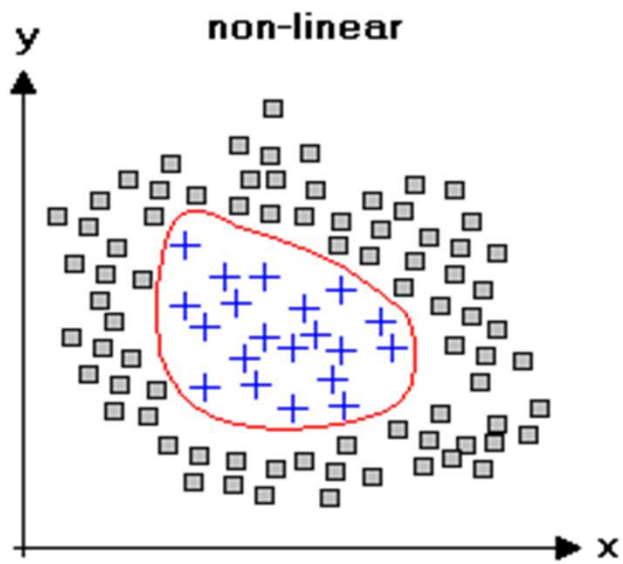
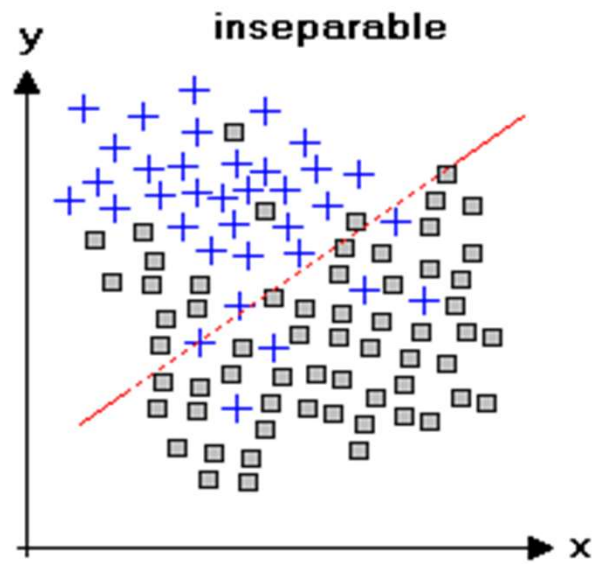
Separability in Feature Space



Linearly Separable



Separable by Non-Linear Boundary



Not Separable in Feature Space

Classification: Input Data & Label

$$\boldsymbol{\omega_0} : \boldsymbol{D_0} = \{\boldsymbol{x}_i^0 : \boldsymbol{x}_i^0 \in \mathbb{R}^d; i = 1, \dots n_0\}$$

$$\boldsymbol{\omega_1} : \boldsymbol{D_1} = \{\boldsymbol{x}_j^1 : \boldsymbol{x}_j^1 \in \mathbb{R}^d; j = 1, \dots n_1\}$$

$$y(\boldsymbol{x}) = \begin{cases} 1, & \boldsymbol{x} \in \boldsymbol{\omega_1} \\ 0, & \boldsymbol{x} \in \boldsymbol{\omega_0} \end{cases}$$

Classification: Input Distribution

$$\boldsymbol{x} \in \boldsymbol{\omega_0} \Rightarrow \boldsymbol{x} \sim P_0(\boldsymbol{x}; \boldsymbol{\theta}_0)$$

$$\boldsymbol{x} \in \boldsymbol{\omega_1} \Rightarrow \boldsymbol{x} \sim P_1(\boldsymbol{x}; \boldsymbol{\theta}_1)$$

P_0 and P_1 are the respective Probability Distributions learned from $\boldsymbol{D_0}$ and $\boldsymbol{D_1}$. The respective parameters of these Distributions are $\boldsymbol{\theta}_0$ and $\boldsymbol{\theta_1}$.

Classification: Input Distribution

$$P(\boldsymbol{\omega_1}|\boldsymbol{x}) = \frac{P(\boldsymbol{x}|\boldsymbol{\omega_1}, \boldsymbol{\theta_1})P(\boldsymbol{\omega_1})}{P(\boldsymbol{x})}$$

Posterior **Likelihood** **Prior**

$$P(\boldsymbol{\omega_0}|\boldsymbol{x}) = \frac{P(\boldsymbol{x}|\boldsymbol{\omega_0}, \boldsymbol{\theta_0})P(\boldsymbol{\omega_0})}{P(\boldsymbol{x})}$$

Evidence

$$P(\boldsymbol{x}) = P(\boldsymbol{x}|\boldsymbol{\omega_0}, \boldsymbol{\theta_0})P(\boldsymbol{\omega_0}) + P(\boldsymbol{x}|\boldsymbol{\omega_1}, \boldsymbol{\theta_1})P(\boldsymbol{\omega_1})$$

Discriminant Functions & Decision Rule

MAP Decision Rule

$$y(\boldsymbol{x}) = \begin{cases} 1, & P(\boldsymbol{\omega_1}|\boldsymbol{x}) > P(\boldsymbol{\omega_0}|\boldsymbol{x}) \\ 0, & P(\boldsymbol{\omega_1}|\boldsymbol{x}) < P(\boldsymbol{\omega_0}|\boldsymbol{x}) \end{cases}$$

Discriminant Function

$$g_i(\boldsymbol{x}) = \ln\{P(\boldsymbol{\omega_i}|\boldsymbol{x})\}$$

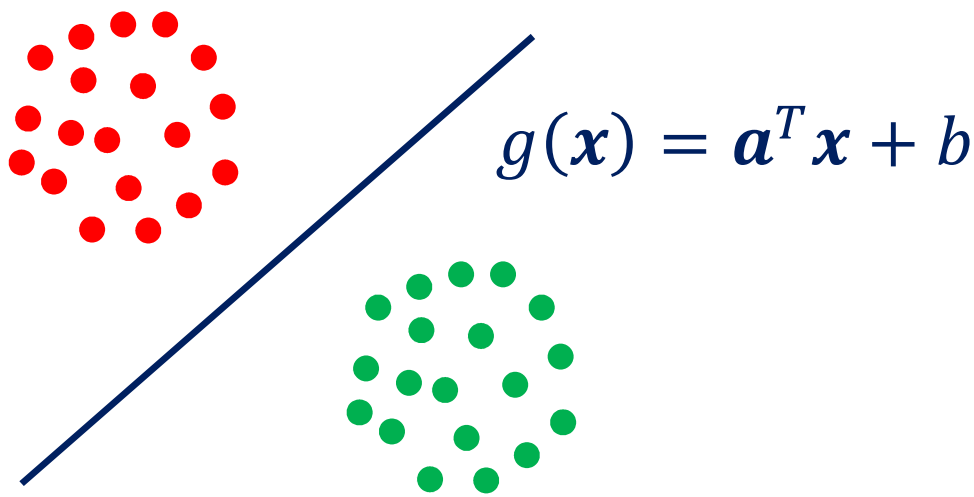
$$g(\boldsymbol{x}) = g_1(\boldsymbol{x}) - g_0(\boldsymbol{x})$$

$$\boxed{y(\boldsymbol{v}) = \begin{cases} 1, & g(\boldsymbol{v}) = g_1(\boldsymbol{v}) - g_0(\boldsymbol{v}) > 0 \\ 0, & g(\boldsymbol{v}) = g_1(\boldsymbol{v}) - g_0(\boldsymbol{v}) < 0 \end{cases}}$$

Classification Decision Rule (unseen data $\boldsymbol{v}$)

Discriminant Functions: Case - I

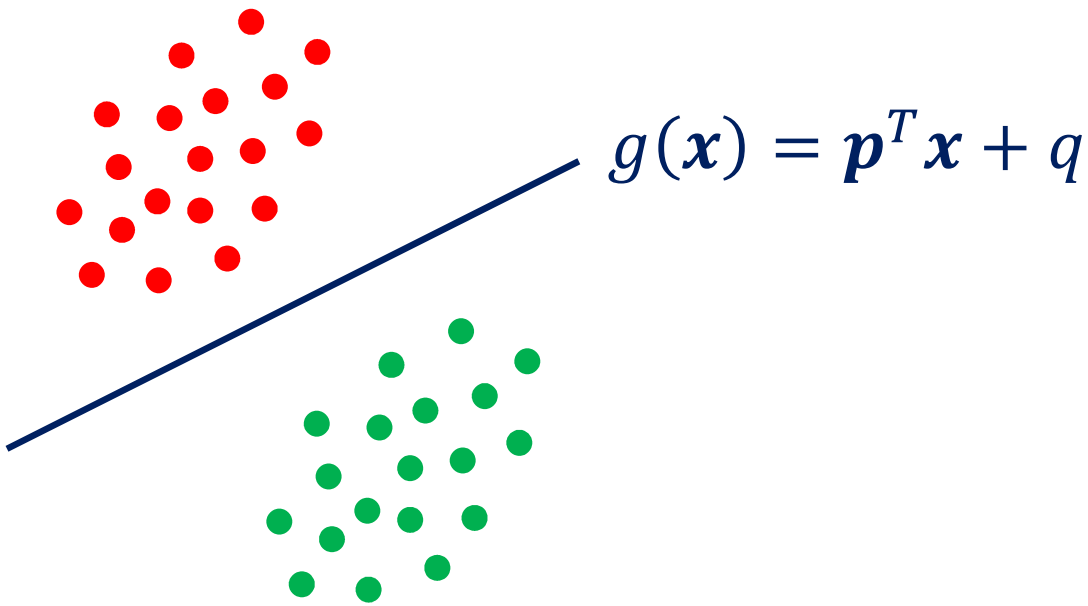
$C_0 = C_1 = \sigma^2 I$



Instances from both Classes are Normally Distributed

Discriminant Functions: Case - II

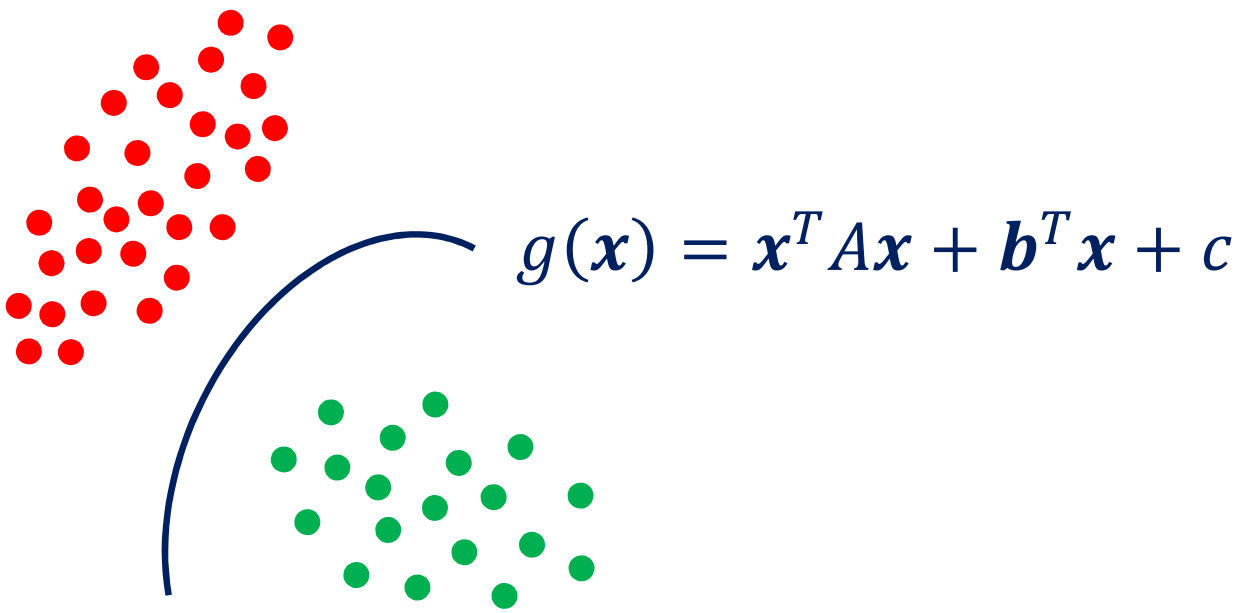
$C_0 = C_1 = C$



Instances from both Classes are Normally Distributed

Discriminant Functions: Case - II

$C_0 \neq C_1$



Instances from both Classes are Normally Distributed

Log Odds and Logit Transform

$y = P(\omega_1 \mid \boldsymbol{x}) \Rightarrow 1 - y = P(\omega_0 \mid \boldsymbol{x})$

Discriminant Function: $g(\boldsymbol{x}) = log\left\{\frac{P(\omega_1|\boldsymbol{x})}{P(\omega_0|\boldsymbol{x})}\right\} = log\left\{\frac{y}{1-y}\right\}$

No Assumption on the Distribution of Instances from $\boldsymbol{D_0}$ and $\boldsymbol{D_1}$

$$g(\boldsymbol{x}) = \boldsymbol{w}^T\boldsymbol{x} + w_0$$

Assumption: $\boldsymbol{g}(\boldsymbol{x})$ is Linear

$y = h(\boldsymbol{x}) = ?$

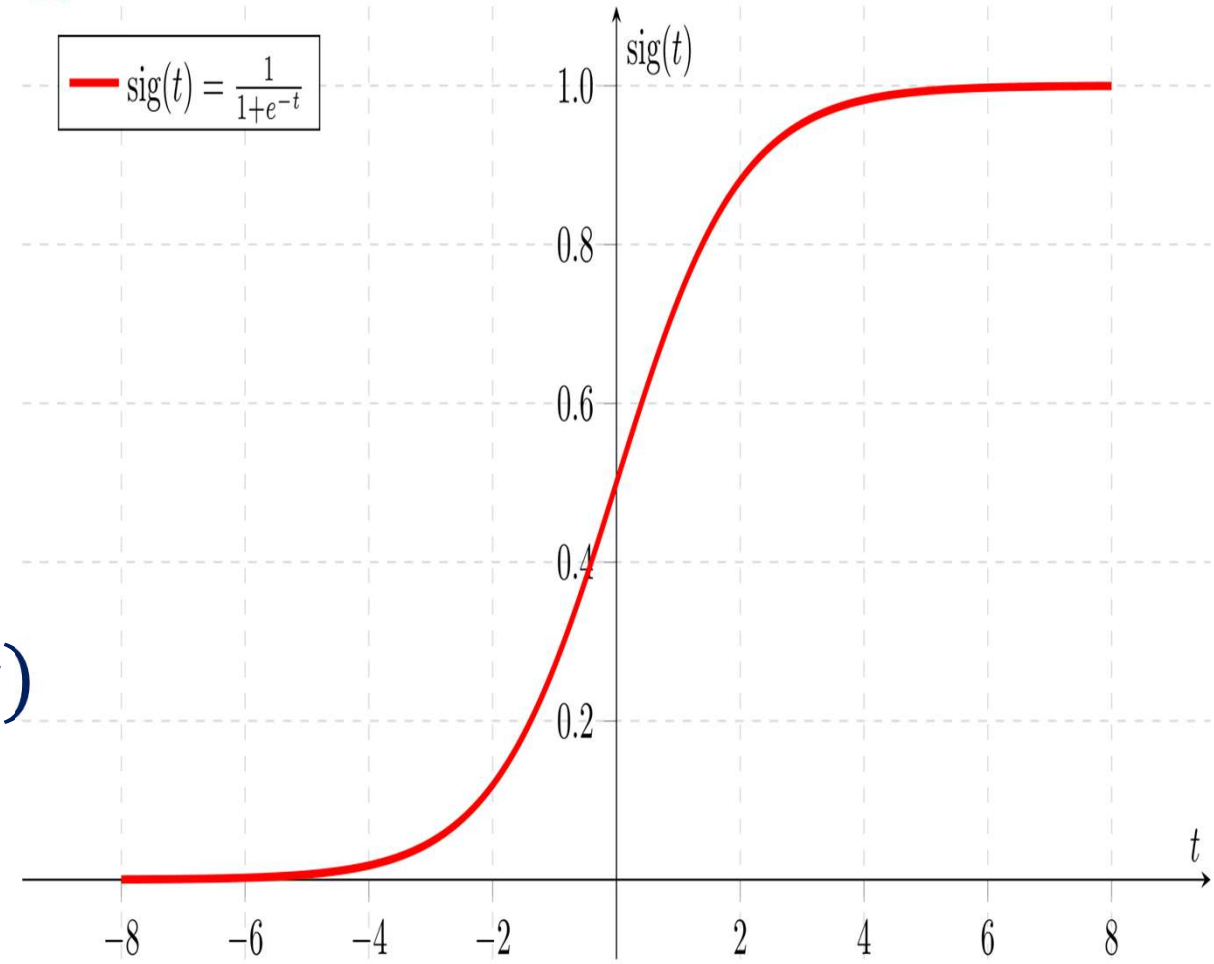
The Log Odds and Sigmoid

$$log\left\{\frac{y}{1-y}\right\} = g(\boldsymbol{x})$$

$$\frac{1-y}{y} = \mathrm{e}^{-g(\boldsymbol{x})}$$

$$y = \frac{1}{1+\mathrm{e}^{-g(\boldsymbol{x})}} = h(\boldsymbol{x})$$

$$g(\boldsymbol{x}) = \boldsymbol{w}^T\boldsymbol{x} + \mathrm{w}_0$$



The Likelihood Function

$$P(y = 1 \mid \boldsymbol{x}) = h(\boldsymbol{x}) \qquad P(y = 0 \mid \boldsymbol{x}) = 1 - h(\boldsymbol{x})$$

$\boldsymbol{x} \in \boldsymbol{\omega_1}$ $\qquad$ $\boldsymbol{x} \in \boldsymbol{\omega_0}$

$$P(y \mid \boldsymbol{x}) = \{h(\boldsymbol{x})\}^{y}\{1 - h(\boldsymbol{x})\}^{1-y}$$

The Log-Likelihood

$$P(\, y \mid \boldsymbol{x}; \boldsymbol{w}, w_0\,) = \{h(\boldsymbol{x})\}^{y}\{1 - h(\boldsymbol{x})\}^{1-y}$$

$$\boldsymbol{S} = \{(\boldsymbol{x}_i, y_i); i = 1, \ldots n\}$$

i.i.d.

$$l(\boldsymbol{w}, w_0) = \prod_{i=1}^{n} P(\, y_i \mid \boldsymbol{x}_i; \boldsymbol{w}, w_0\,)$$

The Log-Likelihood

$$l(\boldsymbol{w}, w_0) = \prod_{i=1}^{n} \{h(\boldsymbol{x}_i)\}^{y_i} \{1 - h(\boldsymbol{x}_i)\}^{1-y_i} \uparrow$$

Log Likelihood

$$\mathrm{L}(\boldsymbol{w}, w_0) = \sum_{i=1}^{n} [y_i log\{h(\boldsymbol{x}_i)\} + \{1 - y_i\} log\{1 - h(\boldsymbol{x}_i)\}] \uparrow$$

Observation

$$\mathrm{L}(\boldsymbol{w}, w_0) = \sum_{i=1}^{n} [y_i log\{h(\boldsymbol{x}_i)\} + \{1 - y_i\} log\{1 - h(\boldsymbol{x}_i)\}] \uparrow$$

$$\downarrow$$

$$\mathrm{L}_{BCE} = -\sum_{i=1}^{n} [y_i log\{h(\boldsymbol{x}_i)\} + \{1 - y_i\} log\{1 - h(\boldsymbol{x}_i)\}] \downarrow$$

Binary Cross Entropy Loss

Maximizing the Log-Likelihood

$$\boldsymbol{w}_{\mathrm{j}}^{(\mathrm{k}+1)} = \boldsymbol{w}_{j}^{(k)} + \eta_{jk} \frac{\partial \mathrm{L}(\boldsymbol{w}, w_0)}{\partial \boldsymbol{w}_j}$$

$j = 1, \ldots d$

Gradient Ascent

$$w_0^{(\mathrm{k}+1)} = w_0^{(k)} + \eta_k \frac{\partial \mathrm{L}(\boldsymbol{w}, w_0)}{\partial w_0}$$

Evaluating the Gradient

$$\frac{\partial \mathrm{L}(\boldsymbol{w}, w_0)}{\partial \boldsymbol{w}_j} = ?$$

$j = 1, \dots d$

$$\frac{\partial \mathrm{L}(\boldsymbol{w}, w_0)}{\partial \boldsymbol{w}_0} = ?$$

Maximizing the Log-Likelihood

$$\mathrm{L}(\boldsymbol{w}, w_0) = \sum_{i=1}^{n} [y_i log\{h(\boldsymbol{x}_i)\} + \{1 - y_i\} log\{1 - h(\boldsymbol{x}_i)\}]$$

$$= \sum_{i=1}^{n} [y_i log\{h(\boldsymbol{x}_i)\} + log\{1 - h(\boldsymbol{x}_i)\} - y_i log\{1 - h(\boldsymbol{x}_i)\}]$$

$$= \sum_{i=1}^{n} \left[y_i log \left\{ \frac{h(\boldsymbol{x}_i)}{1 - h(\boldsymbol{x}_i)} \right\} + log\{1 - h(\boldsymbol{x}_i)\} \right]$$

Maximizing the Log-Likelihood

$$\mathrm{L}(\boldsymbol{w}, w_0) = \sum_{i=1}^{n} \left[y_i log \left\{ \frac{h(\boldsymbol{x}_i)}{1 - h(\boldsymbol{x}_i)} \right\} + log\{1 - h(\boldsymbol{x}_i)\} \right]$$

$$h(\boldsymbol{x}_i) = \frac{1}{1 + e^{-u_i}} \Rightarrow \frac{h(\boldsymbol{x}_i)}{1 - h(\boldsymbol{x}_i)} = e^{u_i}$$

$$\boxed{u_i = g(\boldsymbol{x}_i) = \sum_{r=1}^{d} \boldsymbol{w}[r]\boldsymbol{x}_i[r] + w_0}$$

$$\mathrm{L}(\boldsymbol{w}, w_0) = \sum_{i=1}^{n} [y_i u_i + log\{1 - h(x_i)\}]$$

Maximizing the Log-Likelihood

$$\mathrm{L}(\boldsymbol{w}, w_0) = \sum_{i=1}^{n} [y_i u_i + log\{1 - h(x_i)\}]$$

$$\Downarrow$$

$$\frac{\partial \mathrm{L}(\boldsymbol{w}, w_0)}{\partial \boldsymbol{w}_j} = \sum_{i=1}^{n} \left[y_i \frac{\partial u_i}{\partial \boldsymbol{w}_j} - \frac{1}{1 - h(\boldsymbol{x}_i)} \frac{\partial h(\boldsymbol{x}_i)}{\partial \boldsymbol{w}_j} \right]$$

Maximizing the Log-Likelihood

$$u_i = \sum_{r=1}^{d} \boldsymbol{w}[r]\boldsymbol{x}_i[r] + w_0 \quad \Rightarrow \quad \frac{\partial u_i}{\partial \boldsymbol{w}_j} = \boldsymbol{x}_i[j]$$

$$h(\boldsymbol{x}_i) = \frac{1}{1 + e^{-u_i}}$$

$$\frac{\partial h(\boldsymbol{x}_i)}{\partial \boldsymbol{w}_j} = \frac{\partial h(\boldsymbol{x}_i)}{\partial u_i} \times \frac{\partial u_i}{\partial \boldsymbol{w}_j} = -(1 + e^{-u_i})^{-2}(-e^{-u_i})(\boldsymbol{x}_i[j])$$

$$\frac{\partial h(\boldsymbol{x}_i)}{\partial \boldsymbol{w}_j} = \frac{e^{-u_i}}{(1 + e^{-u_i})^2}(\boldsymbol{x}_i[j]) = h(\boldsymbol{x}_i)\{1 - h(\boldsymbol{x}_i)\}(\boldsymbol{x}_i[j])$$

Maximizing the Log-Likelihood

$$\frac{\partial \mathrm{L}(\boldsymbol{w}, w_0)}{\partial \boldsymbol{w}_j} = \sum_{i=1}^{n} \left[y_i \frac{\partial u_i}{\partial \boldsymbol{w}_j} - \frac{1}{1 - h(\boldsymbol{x}_i)} \frac{\partial h(\boldsymbol{x}_i)}{\partial \boldsymbol{w}_j} \right]$$

$$\Downarrow$$

$$\frac{\partial \mathrm{L}(\boldsymbol{w}, w_0)}{\partial \boldsymbol{w}_j} = \sum_{i=1}^{n} \left[y_i \boldsymbol{x}_i[j] - \frac{1}{1 - h(\boldsymbol{x}_i)} h(\boldsymbol{x}_i)\{1 - h(\boldsymbol{x}_i)\}(\boldsymbol{x}_i[j]) \right]$$

Maximizing the Log-Likelihood

$$\frac{\partial \mathrm{L}(\boldsymbol{w}, w_0)}{\partial \boldsymbol{w}_j} = \sum_{i=1}^{n} \{y_i - h(\boldsymbol{x}_i)\}(\boldsymbol{x}_i[j])$$

$$\frac{\partial \mathrm{L}(\boldsymbol{w}, w_0)}{\partial w_0} = \sum_{i=1}^{n} \{y_i - h(\boldsymbol{x}_i)\}$$

- Random Initialization of Weights
- Dimension-wise Normalization of Input Features

Summary

- Discriminant Function as *Log of Ratio of Posteriors*
- No Assumption on Distribution Function of Instances in Classes
- Instances in Training Datasets of Each Class are *i.i.d.*
- Logistic Regression: Direct Prediction of Posterior from Feature
- Assume a Linear Model for Log of Ratio of Posteriors
- Maximum Likelihood Formulation
- Objective: Maximization of the Log Likelihood
- Genesis of Binary Cross Entropy Loss Function
- Genesis of Sigmoid Function
- Application: Cardiac Arrest Risk Prediction by Logistic Regression

Thank You